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1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing

```
hasini@DESKTOP-UTCBO51: ~  
hasini@DESKTOP-UTCBO51:~$ gcc --version  
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0  
Copyright (C) 2025 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
  
hasini@DESKTOP-UTCBO51:~$ mkdir OSSP  
hasini@DESKTOP-UTCBO51:~$ cd OSSp  
-bash: cd: OSSp: No such file or directory  
hasini@DESKTOP-UTCBO51:~$ cd OSSP  
hasini@DESKTOP-UTCBO51:~/OSSP$ pwd  
/home/hasini/OSSP  
hasini@DESKTOP-UTCBO51:~/OSSP$ mkdir src include bin  
hasini@DESKTOP-UTCBO51:~/OSSP$ ls  
bin  include  src  
hasini@DESKTOP-UTCBO51:~/OSSP$ cd src  
hasini@DESKTOP-UTCBO51:~/OSSP/src$ pwd  
/home/hasini/OSSP/src
```

```
hasini@DESKTOP-UTCBO51: ~ × + v
GNU nano 8.7.1 fork.c *
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("Before fork()\n");

    fork();

    printf("After fork()\n");

    return 0;
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

```
/home/hasini/OSSP/src
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano fork.c
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc fork.c -o../bin/fork
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/fork
Before fork()
After fork()
After fork()
```

```
After fork()
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano process.c
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc process.c -o ../bin/process
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/process
Process ID: 666
Parent Process ID: 338
```



[illegible]

```
hasini@DESKTOP-UTCBOQ51: ~  
pread64(3, "\\6\\0\\0\\4\\0\\0\\0@\\0\\0\\0\\0\\0\\0@\\0\\0\\0\\0\\0\\0@\\0\\0\\0\\0\\0\\0"..  
. , 840, 64) = 840  
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0  
pread64(3, "\\6\\0\\0\\4\\0\\0\\0@\\0\\0\\0\\0\\0\\0@\\0\\0\\0\\0\\0\\0@\\0\\0\\0\\0\\0\\0"..  
. , 840, 64) = 840  
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7974d3e0  
0000  
mmap(0x7974d3e28000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP  
_DENYWRITE, 3, 0x28000) = 0x7974d3e28000  
mmap(0x7974d3fc0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,  
3, 0x1c0000) = 0x7974d3fc0000  
mmap(0x7974d400e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_  
_DENYWRITE, 3, 0x20d000) = 0x7974d400e000  
mmap(0x7974d4014000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_  
_ANONYMOUS, -1, 0) = 0x7974d4014000  
close(3) = 0  
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =  
0x7974d41d7000  
arch_prctl(ARCH_SET_FS, 0x7974d41d7740) = 0  
set_tid_address(0x7974d41d7d68) = 720  
set_robust_list(0x7974d41d7a20, 24) = 0  
rseq(0x7974d41d76a0, 0x20, 0, 0x53053053) = 0  
mprotect(0x7974d400e000, 16384, PROT_READ) = 0  
mprotect(0x5b01fecae000, 4096, PROT_READ) = 0  
mprotect(0x7974d4224000, 8192, PROT_READ) = 0  
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINI  
TY}) = 0  
getrandom("\\x3b\\xe5\\x55\\x1e\\x34\\x95\\x00\\x51", 8, GRND_NONBLOCK) = 8  
munmap(0x7974d41da000, 20087) = 0  
getpid() = 720  
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0  
brk(NULL) = 0x5b021190f000  
brk(0x5b0211930000) = 0x5b0211930000  
write(1, "Process ID: 720\\n", 16Process ID: 720  
) = 16  
getppid() = 718  
write(1, "Parent Process ID: 718\\n", 23Parent Process ID: 718  
) = 23  
exit_group(0) = ?  
+++ exited with 0 +++  
hasini@DESKTOP-UTCBOQ51:~/OSSP/src$
```

2.



```
GNU nano 8.7.1 input.c *
#include <stdio.h>
#include <termios.h>
#include <unistd.h>

int main() {
    char input[100];
    int i = 0;
    char ch;

    printf("myshell> ");

    struct termios oldt, newt;
    tcgetattr(STDIN_FILENO, &oldt);
    newt = oldt;
    newt.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSANOW, &newt);

    while (1) {
        ch = getchar();

        if (ch == '\n') {
            input[i] = '\0';

            printf("\nYou entered: %s\n", input);
            printf("myshell> ");
            i = 0;
        }
        else if (ch == 127) {
            if (i > 0) {
                i--;
                printf("\b \b");
            }
        }
        else if (i < 99) {
            input[i++] = ch;
            printf("%c", ch);
        }
    }
}
```



```
GNU nano 8.7.1 input.c *
char ch;

printf("myshell> ");

struct termios oldt, newt;
tcgetattr(STDIN_FILENO, &oldt);
newt = oldt;
newt.c_lflag &= ~(ICANON | ECHO);
tcsetattr(STDIN_FILENO, TCSANOW, &newt);

while (1) {
    ch = getchar();

    if (ch == '\n') {
        input[i] = '\0';

        printf("\nYou entered: %s\n", input);
        printf("myshell> ");
        i = 0;
    }
    else if (ch == 127) {
        if (i > 0) {
            i--;
            printf("\b \b");
        }
    }
    else if (i < 99) {
        input[i++] = ch;
        printf("%c", ch);
    }
}

tcsetattr(STDIN_FILENO, TCSANOW, &oldt);

return 0;
}
```

```
hasini@DESKTOP-UTCQ51: ~$  
hasini@DESKTOP-UTCQ51:~$ cd ~/OSSP/src  
hasini@DESKTOP-UTCQ51:~/OSSP/src$ pwd  
/home/hasini/OSSP/src  
hasini@DESKTOP-UTCQ51:~/OSSP/src$ nano input.c  
hasini@DESKTOP-UTCQ51:~/OSSP/src$ gcc input.c -o ../bin/input  
hasini@DESKTOP-UTCQ51:~/OSSP/src$ ../bin/input  
myshell> hello  
You entered: hello  
myshell> helllo  
You entered: helllo  
myshell> hello world  
You entered: hello world  
myshell> hel  
You entered: hel  
myshell> |
```

3.

```
hasini@DESKTOP-UTCQ51: ~$  
hasini@DESKTOP-UTCQ51:~/OSSP/src$ pwd  
/home/hasini/OSSP/src  
hasini@DESKTOP-UTCQ51:~/OSSP/src$ nano history.c  
hasini@DESKTOP-UTCQ51:~/OSSP/src$ |
```

```
#define MAX_HISTORY 10

char *history[MAX_HISTORY];
int history_count = 0;
int history_pos = 0;

void add_history(char *command) {
    if (strlen(command) == 0)
        return;

    if (history_count < MAX_HISTORY) {
        history[history_count] = malloc(strlen(command) + 1);
        strcpy(history[history_count], command);
        history_count++;
    }
}

void show_history_command() {
    if (history_pos >= 0 && history_pos < history_count)
        printf("\r\033[Kmyshell> %s", history[history_pos]);
}

int main() {
    char input[100];
    int i = 0;
    char ch;

    struct termios oldt, newt;

    tcgetattr(STDIN_FILENO, &oldt);
    newt = oldt;
    newt.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSANOW, &newt);

    while (1) {
        printf("\nmyshell> ");
```

**^G** Help  
**^X** Exit

**^O** Write Out  
**^R** Read File

**^F** Where Is  
**^\_\** Replace

**^K** Cut  
**^U** Paste

**^T** Execute  
**^J** Justify

```
hasini@DESKTOP-UTC8Q51: ~ × + v
GNU nano 8.7.1 history.c *
newt.c_lflag &= ~(ICANON | ECHO);
tcsetattr(STDIN_FILENO, TCSANOW, &newt);

while (1) {
    printf("\nmysHELL> ");
    fflush(stdout);

    i = 0;

    while (1) {
        ch = getchar();

        if (ch == '\n') {
            input[i] = '\0';

            if (strcmp(input, "exit") == 0)
                break;

            printf("\nYou entered: %s\n", input);

            add_history(input);
            history_pos = history_count;

            break;
        }

        else if (ch == 127) {
            if (i > 0) {
                i--;
                printf("\b \b");
                fflush(stdout);
            }
        }

        else if (ch == 27) {
            char ch1 = getchar();
            char ch2 = getchar();
        }
    }
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

hasini@DESKTOP-UTC BQ51: ~ × + ▾

GNU nano 8.7.1memory.c \*

```
#include <stdlib.h>
#include <string.h>

struct Node {
    char data[50];
    struct Node *next;
};

int main() {
    struct Node *head = NULL;
    struct Node *newNode;
    struct Node *temp;

    // Allocate memory
    newNode = malloc(sizeof(struct Node));

    if (newNode == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    strcpy(newNode->data, "Hello");
    newNode->next = NULL;
    head = newNode;

    printf("Data: %s\n", head->data);

    // Release memory
    temp = head;
    free(temp);
    head = NULL;

    printf("Memory released successfully\n");

    return 0;
}
```

^G Help

^X Exit

^O Write Out

^R Read File

^F Where Is

^\_ Replace

^K Cut

^U Paste

^T Execute

^J Justify

```
hasini@DESKTOP-UTCBO51: ~ × + ▾
GNU nano 8.7.1 history.c *
char ch2 = getchar();

if (ch1 == '[' && ch2 == 'A') {
    if (history_count > 0 && history_pos > 0) {
        history_pos--;
        show_history_command();
    }
}

else if (ch1 == '[' && ch2 == 'B') {
    if (history_pos < history_count - 1) {
        history_pos++;
        show_history_command();
    }
}

fflush(stdout);
}

else if (i < 99) {
    input[i++] = ch;
    printf("%c", ch);
    fflush(stdout);
}
}

if (strcmp(input, "exit") == 0)
    break;
}

tcsetattr(STDIN_FILENO, TCSANOW, &oldt);

for (int j = 0; j < history_count; j++)
    free(history[j]);

return 0;
}|
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

```
myshell> hello world
You entered: hello world

myshell> hello world
You entered:

myshell> test
You entered: test

myshell> exit
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano memory.c
```

```
myshell> exit
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano memory.c
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc memory.c -o ../bin/memory
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/memory
Data: Hello
Memory released successfully
hasini@DESKTOP-UTCBO51:~/OSSP/src$ valgrind --leak-check=full ../bin/memory
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind # version 3.26.0, or
sudo apt install valgrind # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
hasini@DESKTOP-UTCBO51:~/OSSP/src$ sudo apt install valgrind
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: maximum 3 incorrect authentication attempts
hasini@DESKTOP-UTCBO51:~/OSSP/src$ hasini13
hasini13: command not found
hasini@DESKTOP-UTCBO51:~/OSSP/src$ |
```

```
hasini@DESKTOP-UTC8Q51: ~ x + v
See 'snap info valgrind' for additional versions.
hasini@DESKTOP-UTC8Q51:~/OSSP/src$ sudo apt install valgrind
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: maximum 3 incorrect authentication attempts
hasini@DESKTOP-UTC8Q51:~/OSSP/src$ hasini13
hasini13: command not found
hasini@DESKTOP-UTC8Q51:~/OSSP/src$ sudo apt install valgrind
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
Installing:
  valgrind

Installing dependencies:
  gdb          libdebuginfod-common  libipt2
  libbabeltrace1 libdebuginfod1t64  libsource-highlight-common
  libc6-dbg    libdw1t64             libsource-highlight4t64

Suggested packages:
  gdb-doc gdbserver valgrind-mpi kcachegrind

Summary:
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 39
  Download size: 27.1 MB
  Space needed: 111 MB / 1024 GB available

Continue? [Y/n] y
Get:1 https://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod-common all 0.194-4 [12.3 kB]
Get:2 https://archive.ubuntu.com/ubuntu resolute/main amd64 libdw1t64 amd64 0.194-4 [285 kB]
Get:3 https://archive.ubuntu.com/ubuntu resolute/main amd64 libbabeltrace1 amd64 1.5.11-5build1 [165 kB]
Get:4 https://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod1t64 amd64 0.194-4 [20.7 kB]
Get:5 https://archive.ubuntu.com/ubuntu resolute/main amd64 libipt2 amd64 2.1.2-3 [49.2 kB]
```

```
hasini@DESKTOP-UTCBO51: ~  
1.2-3 [49.2 kB]  
Get:6 https://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlight-common all 3.1.9-4.3build2 [64.2 kB]  
Get:7 https://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlight4t64 amd64 3.1.9-4.3build2 [273 kB]  
Get:8 https://archive.ubuntu.com/ubuntu resolute/main amd64 gdb amd64 17.1-2ubuntu1 [4173 kB]  
Ign:9 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6-dbg amd64 2.43-2ubuntu2.3  
Get:10 https://archive.ubuntu.com/ubuntu resolute/main amd64 valgrind amd64 1:3.26.0-0ubuntu1 [15.7 MB]  
Err:9 https://security.ubuntu.com/ubuntu resolute-updates/main amd64 libc6-dbg amd64 2.43-2ubuntu2.3  
  404 Not Found [IP: 91.189.91.81 443]  
  404 Not Found [IP: 185.125.190.83 443]  
Fetched 20.7 MB in 13s (1583 kB/s)  
Error: Failed to fetch https://security.ubuntu.com/ubuntu/pool/main/g/glibc/libc6-dbg_2.43-2ubuntu2.3_amd64.deb 404 Not Found [IP: 185.125.190.83 443]  
Error: Unable to fetch some archives, maybe run apt update or try with --fix-missing?  
hasini@DESKTOP-UTCBO51:~/OSSP/src$ sudo apt update  
[sudo: authenticate] Password:  
sudo: Authentication failed, try again.  
[sudo: authenticate] Password:  
Hit:1 https://archive.ubuntu.com/ubuntu resolute InRelease  
Get:2 https://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]  
Get:3 https://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]  
Get:4 https://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [478 kB]  
Get:5 https://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]  
Get:6 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [616 kB]  
Get:7 https://security.ubuntu.com/ubuntu resolute-security/main Translation-en [113 kB]  
Get:8 https://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB]  
Get:9 https://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [173 kB]  
Get:10 https://security.ubuntu.com/ubuntu resolute-security/universe Transla
```

```
hasini@DESKTOP-UTCBO51: ~  
tion-en [54.5 kB]  
Get:11 https://security.ubuntu.com/ubuntu resolute-security/universe amd64 C  
omponents [43.4 kB]  
Get:12 https://security.ubuntu.com/ubuntu resolute-security/restricted amd64  
Packages [399 kB]  
Get:13 https://security.ubuntu.com/ubuntu resolute-security/restricted Trans  
lation-en [77.7 kB]  
Get:14 https://security.ubuntu.com/ubuntu resolute-security/multiverse amd64  
Packages [10.8 kB]  
Get:15 https://security.ubuntu.com/ubuntu resolute-security/multiverse Trans  
lation-en [2844 B]  
Get:16 https://archive.ubuntu.com/ubuntu resolute-updates/main Translation-e  
n [148 kB]  
Get:17 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Compone  
nts [97.7 kB]  
Get:18 https://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Pac  
kages [294 kB]  
Get:19 https://archive.ubuntu.com/ubuntu resolute-updates/universe Translati  
on-en [93.7 kB]  
Get:20 https://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Com  
ponents [188 kB]  
Get:21 https://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 P  
ackages [413 kB]  
Get:22 https://archive.ubuntu.com/ubuntu resolute-updates/restricted Transla  
tion-en [80.0 kB]  
Get:23 https://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 P  
ackages [11.6 kB]  
Get:24 https://archive.ubuntu.com/ubuntu resolute-updates/multiverse Transla  
tion-en [3172 B]  
Get:25 https://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 P  
ackages [3132 B]  
Get:26 https://archive.ubuntu.com/ubuntu resolute-backports/universe Transla  
tion-en [7988 B]  
Get:27 https://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 C  
omponents [1060 B]  
Fetched 3766 kB in 5s (780 kB/s)  
123 packages can be upgraded. Run 'apt list --upgradable' to see them.  
hasini@DESKTOP-UTCBO51:~/OSSP/src$ sudo apt install valgrind  
Upgrading:  
  libc-bin      libc-gconv-modules-extra  libc6-dev  
  libc-dev-bin  libc6                    locales
```

```
hasini@DESKTOP-UTCQBQ51: ~  
libc-bin      libc-gconv-modules-extra  libc6-dev  
libc-dev-bin  libc6                     locales  
  
Installing:  
valgrind  
  
Installing dependencies:  
gdb           libdebuginfod-common  libipt2  
libbabeltrace1 libdebuginfod1t64     libsource-highlight-common  
libc6-dbg     libdw1t64             libsource-highlight4t64  
  
Suggested packages:  
gdb-doc gdbserver valgrind-mpi kcachegrind  
  
Summary:  
Upgrading: 6, Installing: 10, Removing: 0, Not Upgrading: 117  
Download size: 17.1 MB / 37.8 MB  
Space needed: 111 MB / 1023 GB available  
  
Continue? [Y/n] Y  
Get:1 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc-dev  
-bin amd64 2.43-2ubuntu2.4 [23.3 kB]  
Get:2 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6-dev  
amd64 2.43-2ubuntu2.4 [2294 kB]  
Get:3 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc-gco  
nv-modules-extra amd64 2.43-2ubuntu2.4 [1362 kB]  
Get:4 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6 am  
d64 2.43-2ubuntu2.4 [2104 kB]  
Get:5 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc-bin  
amd64 2.43-2ubuntu2.4 [701 kB]  
Get:6 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 locales  
all 2.43-2ubuntu2.4 [4225 kB]  
Get:7 https://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6-db  
g amd64 2.43-2ubuntu2.4 [6406 kB]  
Fetched 17.1 MB in 10s (1717 kB/s)  
Preconfiguring packages ...  
(Reading database ... 42276 files and directories currently installed.)  
Preparing to unpack .../libc-dev-bin_2.43-2ubuntu2.4_amd64.deb ...  
Unpacking libc-dev-bin (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...  
Preparing to unpack .../libc6-dev_2.43-2ubuntu2.4_amd64.deb ...  
Unpacking libc6-dev:amd64 (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
```

```
hasini@DESKTOP-UTCBO51: ~ × + ▾
Preparing to unpack .../libc-gconv-modules-extra_2.43-2ubuntu2.4_amd64.deb .
..
Unpacking libc-gconv-modules-extra:amd64 (2.43-2ubuntu2.4) over (2.43-2ubunt
u2.3) ...
Setting up libc-gconv-modules-extra:amd64 (2.43-2ubuntu2.4) ...
(Reading database ... 42276 files and directories currently installed.)
Preparing to unpack .../libc6_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc6:amd64 (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
Setting up libc6:amd64 (2.43-2ubuntu2.4) ...
(Reading database ... 42276 files and directories currently installed.)
Preparing to unpack .../libc-bin_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc-bin (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
Setting up libc-bin (2.43-2ubuntu2.4) ...
Selecting previously unselected package libdebuginfod-common.
(Reading database ... 42276 files and directories currently installed.)
Preparing to unpack .../00-libdebuginfod-common_0.194-4_all.deb ...
Unpacking libdebuginfod-common (0.194-4) ...
Preparing to unpack .../01-locales_2.43-2ubuntu2.4_all.deb ...
Unpacking locales (2.43-2ubuntu2.4) over (2.43-2ubuntu2.3) ...
Selecting previously unselected package libdw1t64:amd64.
Preparing to unpack .../02-libdw1t64_0.194-4_amd64.deb ...
Unpacking libdw1t64:amd64 (0.194-4) ...
Selecting previously unselected package libbabeltrace1:amd64.
Preparing to unpack .../03-libbabeltrace1_1.5.11-5build1_amd64.deb ...
Unpacking libbabeltrace1:amd64 (1.5.11-5build1) ...
Selecting previously unselected package libdebuginfod1t64:amd64.
Preparing to unpack .../04-libdebuginfod1t64_0.194-4_amd64.deb ...
Unpacking libdebuginfod1t64:amd64 (0.194-4) ...
Selecting previously unselected package libipt2.
Preparing to unpack .../05-libipt2_2.1.2-3_amd64.deb ...
Unpacking libipt2 (2.1.2-3) ...
Selecting previously unselected package libsource-highlight-common.
Preparing to unpack .../06-libsource-highlight-common_3.1.9-4.3build2_all.de
b ...
Unpacking libsource-highlight-common (3.1.9-4.3build2) ...
Selecting previously unselected package libsource-highlight4t64:amd64.
Preparing to unpack .../07-libsource-highlight4t64_3.1.9-4.3build2_amd64.deb
...
Unpacking libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Selecting previously unselected package gdb.
Preparing to unpack .../08-gdb_17.1-2ubuntu1_amd64.deb ...
```

```
hasini@DESKTOP-UTCQ51: ~ × + v
Preparing to unpack .../09-libc6-dbg_2.43-2ubuntu2.4_amd64.deb ...
Unpacking libc6-dbg:amd64 (2.43-2ubuntu2.4) ...
Selecting previously unselected package valgrind.
Preparing to unpack .../10-valgrind_1%3a3.26.0-0ubuntu1_amd64.deb ...
Unpacking valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod-common (0.194-4) ...
Setting up locales (2.43-2ubuntu2.4) ...
Generating locales (this might take a while)...
Generation complete.
Setting up libsource-highlight-common (3.1.9-4.3build2) ...
Setting up libc6-dbg:amd64 (2.43-2ubuntu2.4) ...
Setting up libdw1t64:amd64 (0.194-4) ...
Setting up libipt2 (2.1.2-3) ...
Setting up libbabeltrace1:amd64 (1.5.11-5build1) ...
Setting up libc-dev-bin (2.43-2ubuntu2.4) ...
Setting up valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod1t64:amd64 (0.194-4) ...
Setting up libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Setting up gdb (17.1-2ubuntu1) ...
Setting up libc6-dev:amd64 (2.43-2ubuntu2.4) ...
Processing triggers for systemd (259.5-0ubuntu3.4) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.4) ...
hasini@DESKTOP-UTCQ51:~/OSSP/src$ valgrind --leak-check=full ../bin/memory
==3463== Memcheck, a memory error detector
==3463== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==3463== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==3463== Command: ../bin/memory
==3463==
Data: Hello
Memory released successfully
==3463==
==3463== HEAP SUMMARY:
==3463==     in use at exit: 0 bytes in 0 blocks
==3463==   total heap usage: 2 allocs, 2 frees, 1,088 bytes allocated
==3463==
==3463== All heap blocks were freed -- no leaks are possible
==3463==
==3463== For lists of detected and suppressed errors, rerun with: -s
==3463== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
hasini@DESKTOP-UTCQ51:~/OSSP/src$ |
```

4.

```
hasini@DESKTOP-UTCQ51:~$ nano tokenize.c
hasini@DESKTOP-UTCQ51:~$ gcc tokenize.c -o ../bin/tokenize
/usr/bin/x86_64-linux-gnu-ld.bfd: cannot open output file ../bin/tokenize: No such file or directory
collect2: error: ld returned 1 exit status
hasini@DESKTOP-UTCQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCQ51:~/OSSP/src$ mv ~/tokenize.c .
hasini@DESKTOP-UTCQ51:~/OSSP/src$ pwd
/home/hasini/OSSP/src
hasini@DESKTOP-UTCQ51:~/OSSP/src$ gcc tokenize.c -o ../bin/tokenize
hasini@DESKTOP-UTCQ51:~/OSSP/src$ ../bin/tokenize
Enter a command: ls -l file.txt
Tokens:
ls
-l
file.txt
```

```
hasini@DESKTOP-UTCQ51: ~ x + - □ ×
GNU nano 8.7.1 tokenize.c *
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];
    char *token;

    printf("Enter a command: ");
    fgets(input, sizeof(input), stdin);

    token = strtok(input, " \n");

    printf("Tokens:\n");

    while (token != NULL) {
        printf("%s\n", token);
        token = strtok(NULL, " \n");
    }

    return 0;
}
```

```
^G Help
^X Exit
```

```
^O Write Out
^R Read File
```

```
^F Where Is
^_ Replace
```

```
^K Cut
^U Paste
```

```
^T Execute
^J Justify
```

```
file.txt
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano parse.c
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc parse.c -o ../bin/parse
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/parse
Enter command: ls file.txt
Command: ls
Argument: file.txt
hasini@DESKTOP-UTCBO51:~/OSSP/src$ |
```

hasini@DESKTOP-UTCBO51: ~ × + ▾

GNU nano 8.7.1 parse.c \*

```
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];
    char *command;
    char *argument;

    printf("Enter command: ");
    fgets(input, sizeof(input), stdin);

    command = strtok(input, " \n");
    argument = strtok(NULL, " \n");

    printf("Command: %s\n", command);

    if (argument != NULL) {
        printf("Argument: %s\n", argument);
    }

    return 0;
}|
```

^G Help

^X Exit

^O Write Out

^R Read File

^F Where Is

^\_ Replace

^K Cut

^U Paste

^T Execute

^J Justify

5.

```
hasini@DESKTOP-UTCBO51: ~  
hasini@DESKTOP-UTCBO51:~$ ~/OSSP/src  
-bash: /home/hasini/OSSP/src: Is a directory  
hasini@DESKTOP-UTCBO51:~$ cd ~/OSSP/src  
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano quotes.c  
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc quotes.c -o ../bin/quotes  
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/quotes  
Enter a command: "Hello World"  
You entered: "Hello World"  
hasini@DESKTOP-UTCBO51:~/OSSP/src$  
  
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/quotes  
Enter a command: "Hello world"  
Tokens:  
Hello world  
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ls  
append.c      escape.c      modules.c      process.c      stderr.c  
background.c  fork.c        output.txt     processgroup.c test.sh  
builtin.c     history.c     parse.c        quotes.c       tokenize.c  
builtins2.c   input.c       performance.c  redirect.c     variables.c  
complex.c     jobs.c        pipeline.c     signalhandler.c waitpath.c  
error.txt     memory.c      pipes.c        signals.c  
hasini@DESKTOP-UTCBO51:~/OSSP/src$
```

```
hasini@DESKTOP-UTCBO51: ~ × +
GNU nano 8.7.1 quotes.c *
#include <stdio.h>
#include <string.h>

int main()
{
    char input[200];
    char token[100];
    int i = 0;
    int j = 0;
    char quote = 0;

    printf("Enter a command: ");
    fgets(input, sizeof(input), stdin);

    printf("Tokens:\n");

    while (input[i] != '\0' && input[i] != '\n')
    {
        if (input[i] == '"' || input[i] == '\\')
        {
            if (quote == 0)
            {
                quote = input[i];
            }
            else if (quote == input[i])
            {
                quote = 0;
            }
            else
            {
                token[j++] = input[i];
            }
        }
        else if (input[i] == ' ' && quote == 0)
        {
            if (j > 0)
            {

```

|                |                     |                    |                 |                   |
|----------------|---------------------|--------------------|-----------------|-------------------|
| <b>^G</b> Help | <b>^O</b> Write Out | <b>^F</b> Where Is | <b>^K</b> Cut   | <b>^T</b> Execute |
| <b>^X</b> Exit | <b>^R</b> Read File | <b>^V</b> Replace  | <b>^U</b> Paste | <b>^J</b> Justify |

```
hasini@DESKTOP-UTCQ51: ~ × + v
GNU nano 8.7.1 quotes.c *
    quote = input[i];
    }
    else if (quote == input[i])
    {
        quote = 0;
    }
    else
    {
        token[j++] = input[i];
    }
}
else if (input[i] == ' ' && quote == 0)
{
    if (j > 0)
    {
        token[j] = '\0';
        printf("%s\n", token);
        j = 0;
    }
}
else
{
    token[j++] = input[i];
}

i++;
}

if (j > 0)
{
    token[j] = '\0';
    printf("%s\n", token);
}

return 0;
}
|

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

6.

```
hasini@DESKTOP-UTCQ51: ~ × + v
hasini@DESKTOP-UTCQ51:~$ nano escape.c
hasini@DESKTOP-UTCQ51:~$ |
```

```
hasini@DESKTOP-UTCBO51: ~ × + v
GNU nano 8.7.1 escape.c *
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>

int main() {
    printf("Parent process started\n");

    pid_t pid = fork();

    if (pid == 0) {
        printf("Child process executing ls...\n");
        execlp("ls", "ls", NULL);

        printf("exec failed\n");
        exit(1);
    }
    else {
        printf("Parent process continues\n");
    }

    return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify
```

```
hasini@DESKTOP-UTCBO51:~$ gcc escape.c -o ../bin/escape
/usr/bin/x86_64-linux-gnu-ld.bfd: cannot open output file ../bin/escape: No
such file or directory
collect2: error: ld returned 1 exit status
hasini@DESKTOP-UTCBO51:~$ mv ~/escape.c ~/OSSP/src/
hasini@DESKTOP-UTCBO51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc escape.c -o ../bin/escape
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/escape
Parent process started
Parent process continues
Child process executing ls...
escape.c  history.c  memory.c  process.c  tokenize.c
fork.c    input.c    parse.c   quotes.c
hasini@DESKTOP-UTCBO51:~/OSSP/src$
```

7.

```
hasini@DESKTOP-UTCBO51: ~ × + v
hasini@DESKTOP-UTCBO51:~$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBO51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano waitpath.c
hasini@DESKTOP-UTCBO51:~/OSSP/src$ |
```

```
GNU nano 8.7.1 waitpath.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        printf("Child process running...\n");
        execlp("ls", "ls", NULL);
    }
    else {
        waitpid(pid, NULL, 0);
        printf("Child process finished.\n");
        printf("Parent process continues.\n");
    }

    return 0;
}
```

|      |           |          |       |         |
|------|-----------|----------|-------|---------|
| Help | Write Out | Where Is | Cut   | Execute |
| Exit | Read File | Replace  | Paste | Justify |

```
hasini@DESKTOP-UTCBO51:~$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBO51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano waitpath.c
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc waitpath.c -o ../bin/waitpath
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/waitpath
Child process running...
escape.c history.c memory.c process.c tokenize.c
fork.c input.c parse.c quotes.c waitpath.c
Child process finished.
Parent process continues.
hasini@DESKTOP-UTCBO51:~/OSSP/src$
```

8.

```
hasini@DESKTOP-UTCBO51:~$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBO51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano variables.c
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc variables.c -o ../bin/variables
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/variables
Enter a variable: $HOME
/home/hasini
```

```
hasini@DESKTOP-UTCBO51: ~ × + ▾
GNU nano 8.7.1 variables.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main() {
    char input[100];

    printf("Enter a variable: ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    if (strcmp(input, "$HOME") == 0) {
        printf("%s\n", getenv("HOME"));
    }
    else if (strcmp(input, "$USER") == 0) {
        printf("%s\n", getenv("USER"));
    }
    else {
        printf("Variable not found\n");
    }

    return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify

hasini@DESKTOP-UTCBO51:~/OSSP/src$ $user
hasini@DESKTOP-UTCBO51:~/OSSP/src$ $USER
hasini: command not found
hasini@DESKTOP-UTCBO51:~/OSSP/src$ $USER
hasini: command not found
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ls ../bin
escape  history  memory  process  tokenize  waitpath
fork    input   parse   quotes   variables
hasini@DESKTOP-UTCBO51:~/OSSP/src$ |
```

9.



```
GNU nano 8.7.1 builtin.c *
#include <stdio.h>
#include <unistd.h>
#include <string.h>

int main() {
    char command[100];

    while (1) {
        printf("myshell> ");
        fgets(command, sizeof(command), stdin);

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0) {
            break;
        }

        if (strncmp(command, "cd ", 3) == 0) {
            if (chdir(command + 3) == 0)
                printf("Directory changed successfully\n");
            else
                printf("Directory change failed\n");
        }
        else {
            printf("Unknown command\n");
        }
    }

    return 0;
}
```

**^G** Help      **^O** Write Out      **^F** Where Is      **^K** Cut      **^T** Execute  
**^X** Exit      **^R** Read File      **^\_\** Replace      **^U** Paste      **^J** Justify

```
hasini@DESKTOP-UTCBQ51:~$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ nano builtin.c
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ gcc builtin.c -o ../bin/builtin
hasini@DESKTOP-UTCBQ51:~/OSSP/src$ ../bin/builtin
myshell> cd..
Unknown command
myshell> cd..
Unknown command
myshell> cd ..
Directory changed successfully
myshell> cd src
Directory changed successfully
myshell> exit
hasini@DESKTOP-UTCBQ51:~/OSSP/src$
```

10.

```
hasini@DESKTOP-UTCBQ51:~$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBQ51:~$ ~/OSSP/src~/OSSP/src
-bash: /home/hasini/OSSP/src~/OSSP/src: No such file or directory
hasini@DESKTOP-UTCBQ51:~$ cd ~/OSSP/src
```

```
GNU nano 8.7.1                                builtins2.c *
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <stdlib.h>

int main() {
    char command[100];

    while (1) {
        printf("myshell> ");
        fgets(command, sizeof(command), stdin);

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "pwd") == 0) {
            char path[200];
            getcwd(path, sizeof(path));
            printf("%s\n", path);
        }
        else if (strcmp(command, "exit") == 0) {
            break;
        }
        else if (strncmp(command, "export ", 7) == 0) {
            char *value = command + 7;
            putenv(value);
            printf("Variable exported\n");
        }
        else {
            printf("Unknown command\n");
        }
    }

    return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
hasini@DESKTOP-UTCBO51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano builtins2.c
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc builtins2.c -o ../bin/builtins2
```

```
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/builtins2
myshell> pwd
/home/hasini/OSSP/src
myshell> export NAME=Hasini
Variable exported
myshell> exit
hasini@DESKTOP-UTCBO51:~/OSSP/src$ |
```

11.

```
hasini@DESKTOP-UTCQ51: ~$ cd ~/OSSP/src
hasini@DESKTOP-UTCQ51:~/OSSP/src$ nano pipeline.c
hasini@DESKTOP-UTCQ51:~/OSSP/src$ |
```

```
GNU nano 8.7.1 pipeline.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int fd[2];
    pipe(fd);

    pid_t pid = fork();

    if (pid == 0) {
        close(fd[0]);
        dup2(fd[1], STDOUT_FILENO);
        close(fd[1]);

        execlp("ls", "ls", NULL);
    }
    else {
        close(fd[1]);
        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);

        execlp("wc", "wc", "-l", NULL);
        wait(NULL);
    }

    return 0;
}
```

```
hasini@DESKTOP-UTCQ51:~/OSSP/src$ gcc pipeline.c -o ../bin/pipeline
hasini@DESKTOP-UTCQ51:~/OSSP/src$ ../bin/pipeline
14
hasini@DESKTOP-UTCQ51:~/OSSP/src$ |
```

12.

```
hasini@DESKTOP-UTCBO51: ~$ cd ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBO51: ~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBO51: ~/OSSP/src$ gcc pipes.c -o ../bin/pipes
cc1: fatal error: pipes.c: No such file or directory
compilation terminated.
hasini@DESKTOP-UTCBO51: ~/OSSP/src$ nano pipes.c
```

```
hasini@DESKTOP-UTCBO51: ~/OSSP/src$ nano pipes.c
hasini@DESKTOP-UTCBO51: ~/OSSP/src$ gcc pipes.c -o ../bin/pipes
```

```
GNU nano 8.7.1 pipes.c *
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>

int main() {
    int fd[2];
    char message[] = "Hello from child process";
    char buffer[100];

    pipe(fd);

    pid_t pid = fork();

    if (pid == 0) {
        close(fd[0]);

        write(fd[1], message, strlen(message) + 1);

        close(fd[1]);
    }
    else {
        close(fd[1]);

        read(fd[0], buffer, sizeof(buffer));

        printf("Parent received: %s\n", buffer);

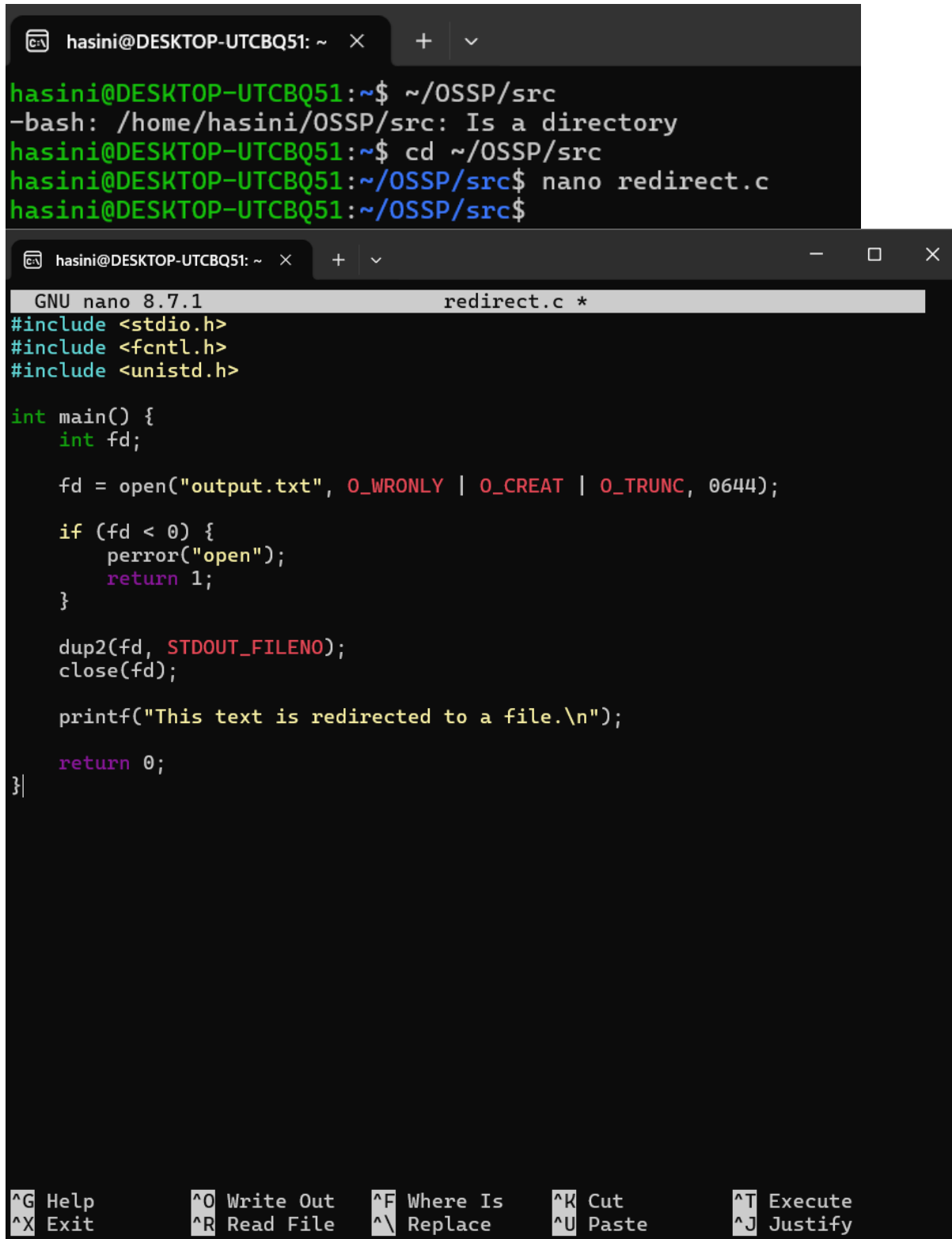
        close(fd[0]);
        wait(NULL);
    }

    return 0;
}
```

|                |                     |                    |                 |                   |
|----------------|---------------------|--------------------|-----------------|-------------------|
| <b>^G</b> Help | <b>^O</b> Write Out | <b>^F</b> Where Is | <b>^K</b> Cut   | <b>^T</b> Execute |
| <b>^X</b> Exit | <b>^R</b> Read File | <b>^N</b> Replace  | <b>^U</b> Paste | <b>^J</b> Justify |

```
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc pipes.c -o ../bin/pipes
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/pipes
Parent received: Hello from child process
hasini@DESKTOP-UTCBO51:~/OSSP/src$
```

13.



```
hasini@DESKTOP-UTCBO51: ~ × + v
hasini@DESKTOP-UTCBO51:~$ ~/OSSP/src
-bash: /home/hasini/OSSP/src: Is a directory
hasini@DESKTOP-UTCBO51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano redirect.c
hasini@DESKTOP-UTCBO51:~/OSSP/src$
```

```
GNU nano 8.7.1 redirect.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0) {
        perror("open");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("This text is redirected to a file.\n");

    return 0;
}
```

|                |                     |                    |                 |                   |
|----------------|---------------------|--------------------|-----------------|-------------------|
| <b>^G</b> Help | <b>^O</b> Write Out | <b>^F</b> Where Is | <b>^K</b> Cut   | <b>^T</b> Execute |
| <b>^X</b> Exit | <b>^R</b> Read File | <b>^N</b> Replace  | <b>^U</b> Paste | <b>^J</b> Justify |

```
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc redirect.c -o ../bin/redirect
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/redirect
hasini@DESKTOP-UTCBO51:~/OSSP/src$ cat output.txt
This text is redirected to a file.
hasini@DESKTOP-UTCBO51:~/OSSP/src$
```

14.

```
hasini@DESKTOP-UTCBO51: ~ × + ▾
hasini@DESKTOP-UTCBO51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano append.c
```

```
GNU nano 8.7.1 append.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("output.txt", O_WRONLY | O_CREAT | O_APPEND, 0644);

    if (fd < 0) {
        perror("open");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("This line is appended.\n");

    return 0;
}
```

```
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc append.c -o ../bin/append
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/append
hasini@DESKTOP-UTCBO51:~/OSSP/src$ cat output.txt
This text is redirected to a file.
This line is appended.
hasini@DESKTOP-UTCBO51:~/OSSP/src$ |

hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano stderr.c
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc stderr.c -o ../bin/stderr
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc stderr.c -o ../bin/stderr
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/stderr
hasini@DESKTOP-UTCBO51:~/OSSP/src$ cat error.txt
This is an error message.
hasini@DESKTOP-UTCBO51:~/OSSP/src$
```

```
GNU nano 8.7.1 stderr.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("error.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0) {
        return 1;
    }

    dup2(fd, STDERR_FILENO);
    close(fd);

    fprintf(stderr, "This is an error message.\n");

    return 0;
}|

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

15.

```
hasini@DESKTOP-UTCBO51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano complex.c
hasini@DESKTOP-UTCBO51:~/OSSP/src$
```

```
hasini@DESKTOP-UTCBO51: ~ × + ▾
hasini@DESKTOP-UTCBO51:~$
hasini@DESKTOP-UTCBO51:~$ #include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char input[200];

    printf("myshell> ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    if (strcmp(input, "exit") == 0)
        return 0;

    char *command1 = strtok(input, "|");
    char *command2 = strtok(NULL, "|");

    if (command2 != NULL)
    {
        int pipefd[2];
        pipe(pipefd);

        pid_t pid1 = fork();

        if (pid1 == 0)
        {
            dup2(pipefd[1], STDOUT_FILENO);
            close(pipefd[0]);
            close(pipefd[1]);

            char *args[10];
            int i = 0;
            char *token = strtok(command1, " ");

            while (token != NULL)
            {
```

```
    while (token != NULL)
    {
        args[i++] = token;
        token = strtok(NULL, " ");
    }

    args[i] = NULL;
    execvp(args[0], args);
    perror("exec failed");
    exit(1);
}

pid_t pid2 = fork();

if (pid2 == 0)
{
    dup2(pipefd[0], STDIN_FILENO);
    close(pipefd[1]);
    close(pipefd[0]);

    char *args[10];
    int i = 0;
    char *token = strtok(command2, " ");

    while (token != NULL)
    {
        args[i++] = token;
        token = strtok(NULL, " ");
    }

    args[i] = NULL;
    execvp(args[0], args);
    perror("exec failed");
    exit(1);
}
```

**^G** Help  
**^X** Exit

**^O** Write Out  
**^R** Read File

**^F** Where Is  
**^\_** Replace

**^K** Cut  
**^U** Paste

**^T** Execute  
**^J** Justify

```
        args[i] = NULL;
        execvp(args[0], args);
        perror("exec failed");
        exit(1);
    }

    pid_t pid2 = fork();

    if (pid2 == 0)
    {
        dup2(pipefd[0], STDIN_FILENO);
        close(pipefd[1]);
        close(pipefd[0]);

        char *args[10];
        int i = 0;
        char *token = strtok(command2, " ");

        while (token != NULL)
        {
            args[i++] = token;
            token = strtok(NULL, " ");
        }

        args[i] = NULL;
        execvp(args[0], args);
        perror("exec failed");
        exit(1);
    }

    close(pipefd[0]);
    close(pipefd[1]);

    waitpid(pid1, NULL, 0);
    waitpid(pid2, NULL, 0);
}
```

**^G** Help  
**^X** Exit

**^O** Write Out  
**^R** Read File

**^F** Where Is  
**^\\** Replace

**^K** Cut  
**^U** Paste

**^T** Execute  
**^J** Justify

```
hasini@DESKTOP-UTCBO51: ~ × + v
hasini@DESKTOP-UTCBO51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano complex.c
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc complex.c -o ../bin/complex
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/complex
myshell> ls
Executing command: ls
append.c      escape.c      modules.c      process.c      stderr.c
background.c  fork.c        output.txt     processgroup.c test.sh
builtin.c     history.c     parse.c        quotes.c        tokenize.c
builtins2.c   input.c       performance.c  redirect.c      variables.c
complex.c     jobs.c        pipeline.c     signalhandler.c waitpath.c
error.txt     memory.c      pipes.c        signals.c
hasini@DESKTOP-UTCBO51:~/OSSP/src$ wc -l
error: unexpected argument '-l' found

    tip: to pass '-l' as a value, use '-- -l'

Usage: wc [OPTION]... [FILE]...

For more information, try '--help'.
hasini@DESKTOP-UTCBO51:~/OSSP/src$ wc -l
^Z
[1]+  Stopped                  wc -l
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ls | wc -l
29
hasini@DESKTOP-UTCBO51:~/OSSP/src$
```

16.



```
GNU nano 8.7.1 background.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

int main()
{
    char input[100];

    printf("myshell> ");
    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';

    int background = 0;
    int len = strlen(input);

    if (len > 0 && input[len - 1] == '&')
    {
        background = 1;
        input[len - 1] = '\0';
    }

    pid_t pid = fork();

    if (pid == 0)
    {
        char *args[10];
        int i = 0;

        char *token = strtok(input, " ");

        while (token != NULL)
        {
            args[i++] = token;
            token = strtok(NULL, " ");
        }

        ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
        ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
hasini@DESKTOP-UTCQ51: ~ × + ▾
GNU nano 8.7.1 background.c *
if (pid == 0)
{
    char *args[10];
    int i = 0;

    char *token = strtok(input, " ");

    while (token != NULL)
    {
        args[i++] = token;
        token = strtok(NULL, " ");
    }

    args[i] = NULL;

    execvp(args[0], args);
    perror("exec failed");
    exit(1);
}
else
{
    printf("Process started. PID: %d\n", pid);

    if (background)
    {
        printf("Running in background.\n");
    }
    else
    {
        waitpid(pid, NULL, 0);
        printf("Foreground process completed.\n");
    }
}

return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
hasini@DESKTOP-UTCQ51: ~ × + ▾
hasini@DESKTOP-UTCQ51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCQ51:~/OSSP/src$ nano background.c
hasini@DESKTOP-UTCQ51:~/OSSP/src$ gcc background.c -o ../bin/background
hasini@DESKTOP-UTCQ51:~/OSSP/src$ ../bin/background
mysHELL> sleep 5 &
Process started. PID: 575
Running in background.
hasini@DESKTOP-UTCQ51:~/OSSP/src$ |
```

17.

```
hasini@DESKTOP-UTCQ51: ~$ cd ~/OSSP/src
hasini@DESKTOP-UTCQ51:~/OSSP/src$ nano jobs.c
```

```
GNU nano 8.7.1 jobs.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

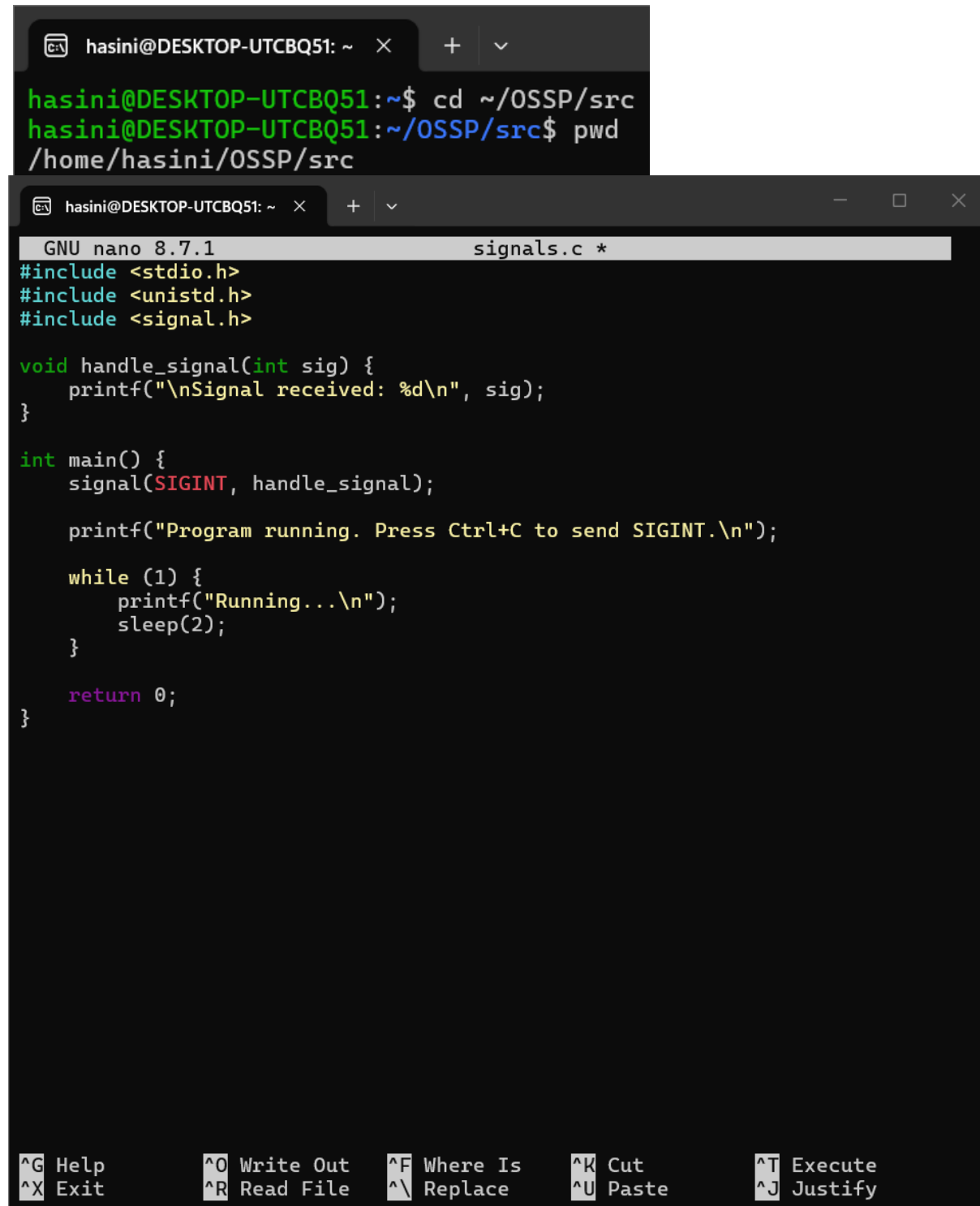
int main() {
    pid_t pid = fork();

    if (pid == 0) {
        printf("Child process running. PID: %d\n", getpid());
        sleep(10);
        printf("Child process completed.\n");
    }
    else {
        printf("Parent process: Child PID = %d\n", pid);
        waitpid(pid, NULL, 0);
        printf("Child returned to foreground.\n");
    }

    return 0;
}
```

```
hasini@DESKTOP-UTCQ51: ~$ cd ~/OSSP/src
hasini@DESKTOP-UTCQ51:~/OSSP/src$ nano jobs.c
hasini@DESKTOP-UTCQ51:~/OSSP/src$ gcc jobs.c -o ../bin/jobs
hasini@DESKTOP-UTCQ51:~/OSSP/src$ ../bin/jobs
Job running in background. PID: 802
Bringing job to foreground...
Background job started. PID: 802
Background job completed.
Job completed in foreground.
hasini@DESKTOP-UTCQ51:~/OSSP/src$
```

18.



The image shows a terminal window with two panes. The top pane shows a shell session where the user navigates to the directory `~/OSSP/src` and confirms the path with `pwd`. The bottom pane shows the GNU nano 8.7.1 text editor editing a file named `signals.c`. The code in the editor is a C program that includes `<stdio.h>`, `<unistd.h>`, and `<signal.h>`. It defines a `handle_signal` function that prints a message when a signal is received. The `main` function sets up a signal handler for `SIGINT`, prints a message, and enters a loop that prints "Running..." and sleeps for 2 seconds.

```
hasini@DESKTOP-UTCBO51: ~  
hasini@DESKTOP-UTCBO51:~$ cd ~/OSSP/src  
hasini@DESKTOP-UTCBO51:~/OSSP/src$ pwd  
/home/hasini/OSSP/src  
  
GNU nano 8.7.1 signals.c *  
#include <stdio.h>  
#include <unistd.h>  
#include <signal.h>  
  
void handle_signal(int sig) {  
    printf("\nSignal received: %d\n", sig);  
}  
  
int main() {  
    signal(SIGINT, handle_signal);  
  
    printf("Program running. Press Ctrl+C to send SIGINT.\n");  
  
    while (1) {  
        printf("Running...\n");  
        sleep(2);  
    }  
  
    return 0;  
}
```

Help Write Out Where Is Cut Execute  
Exit Read File Replace Paste Justify

```
hasini@DESKTOP-UTCBOQ1:~/OSSP/src$ mv ~/signals.c ~/OSSP/src/  
hasini@DESKTOP-UTCBOQ1:~/OSSP/src$ gcc signals.c -o ../bin/signals  
hasini@DESKTOP-UTCBOQ1:~/OSSP/src$ ../bin/signals  
Program running. Press Ctrl+C to send SIGINT.  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
Running...  
../bin/signalsRunning...  
Running...  
^C  
Signal received: 2  
Running...  
Running...  
Running...  
^C  
Signal received: 2  
Running...  
Running...  
^C  
Signal received: 2  
Running...  
Running...
```

```
^C
Signal received: 2
Running...
Running...
Running...
Running...
Running...
Running...
```

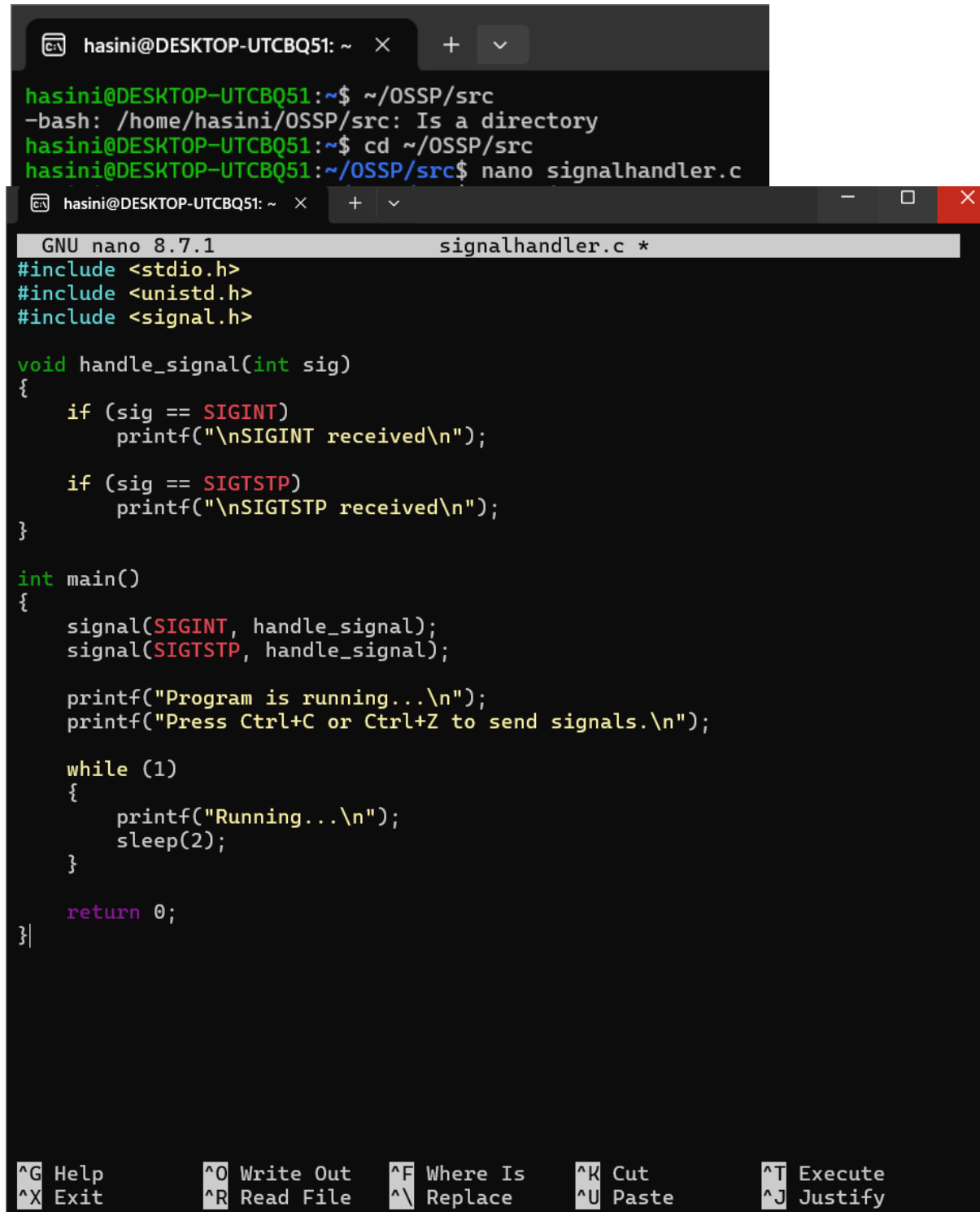
```
^C
Signal received: 2
Running...
Running...
Running...
Running...
Running...
Running...
```

```
^C
Signal received: 2
Running...
Running...
^C
```

[illegible]

```
^Z
[1]+  Stopped                  ../bin/signals
hasini@DESKTOP-UTCBO51:~/OSSP/src$
```

19.



The image shows a terminal window with two panes. The top pane shows a user named 'hasini' at a machine named 'DESKTOP-UTCQBQ51'. The user enters the command `~/OSSP/src`, which results in a message: `-bash: /home/hasini/OSSP/src: Is a directory`. Then, the user enters `cd ~/OSSP/src`, and finally `nano signalhandler.c`. The bottom pane shows the nano text editor editing the file `signalhandler.c`. The code in the file is as follows:

```
GNU nano 8.7.1 signalhandler.c *
#include <stdio.h>
#include <unistd.h>
#include <signal.h>

void handle_signal(int sig)
{
    if (sig == SIGINT)
        printf("\nSIGINT received\n");

    if (sig == SIGTSTP)
        printf("\nSIGTSTP received\n");
}

int main()
{
    signal(SIGINT, handle_signal);
    signal(SIGTSTP, handle_signal);

    printf("Program is running...\n");
    printf("Press Ctrl+C or Ctrl+Z to send signals.\n");

    while (1)
    {
        printf("Running...\n");
        sleep(2);
    }

    return 0;
}
```

At the bottom of the terminal window, there is a status bar with the following keyboard shortcuts and their functions:

|                      |                           |                          |                       |                         |
|----------------------|---------------------------|--------------------------|-----------------------|-------------------------|
| <code>^G</code> Help | <code>^O</code> Write Out | <code>^F</code> Where Is | <code>^K</code> Cut   | <code>^T</code> Execute |
| <code>^X</code> Exit | <code>^R</code> Read File | <code>^_\</code> Replace | <code>^U</code> Paste | <code>^J</code> Justify |

```
hasini@DESKTOP-UTC8Q51:~/OSSP/src$ gcc signalhandler.c -o ../bin/signalhandler
hasini@DESKTOP-UTC8Q51:~/OSSP/src$ ./bin/signalhandler
Program is running...
Press Ctrl+C or Ctrl+Z to send signals.
Running...
Running...
Running...
^Z
SIGTSTP received
Running...
Running...
Running...
Running...
^Z
SIGTSTP received
Running...
Running...
Running...
Running...
Running...
Running...
^Z
SIGTSTP received
Running...
^XRunning...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
killRunning...
Running...
51
Running...
Running...
killRunning...
Running...
Running...
Running...
```



hasini@DESKTOP-UTCQB51: ~ ×

+

[illegible]

```
Running...
Running...
Running...
Running...
Running...
Running...
Running...
^\\Quit                                     ../bin/signalhandler
hasini@DESKTOP-UTCBO51:~/OSSP/src$
```

20.

```
hasini@DESKTOP-UTCBO51: ~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano processgroup.c
```

```
GNU nano 8.7.1 processgroup.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>

int main()
{
    printf("Process ID: %d\n", getpid());
    printf("Process Group ID: %d\n", getpgrp());
    printf("Parent Process ID: %d\n", getppid());

    return 0;
}
```

```
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

```
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc processgroup.c -o ../bin/processgroup
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/processgroup
Process ID: 791
Process Group ID: 791
Parent Process ID: 672
hasini@DESKTOP-UTCBO51:~/OSSP/src$
```

21.

```
hasini@DESKTOP-UTCBO51: ~  
hasini@DESKTOP-UTCBO51:~$ cd~/OSSP/src  
-bash: cd~/OSSP/src: No such file or directory  
hasini@DESKTOP-UTCBO51:~$ cd ~/OSSP/src  
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano modules.c  
GNU nano 8.7.1 modules.c *  
#include <stdio.h>  
#include <stdlib.h>  
  
int main()  
{  
    FILE *file;  
  
    file = fopen("test.txt", "r");  
  
    if (file == NULL)  
    {  
        perror("Error opening file");  
        return 1;  
    }  
  
    printf("File opened successfully.\n");  
  
    fclose(file);  
  
    return 0;  
}  
  
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify  
  
hasini@DESKTOP-UTCBO51:~/OSSP/src$ gcc modules.c -o ../bin/modules  
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ../bin/modules  
Error opening file: No such file or directory  
hasini@DESKTOP-UTCBO51:~/OSSP/src$ |
```

## 22. Valgrind and Test Scripts:

```
hasini@DESKTOP-UTCBO51:~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBO51:~/OSSP/src$ valgrind --leak-check=full ../bin/memory
==827== Memcheck, a memory error detector
==827== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==827== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==827== Command: ../bin/memory
==827==
Data: Hello
Memory released successfully
==827==
==827== HEAP SUMMARY:
==827==     in use at exit: 0 bytes in 0 blocks
==827==   total heap usage: 2 allocs, 2 frees, 1,088 bytes allocated
==827==
==827== All heap blocks were freed -- no leaks are possible
==827==
==827== For lists of detected and suppressed errors, rerun with: -s
==827== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

```
hasini@DESKTOP-UTCBO51:~/OSSP/src$ nano test.sh
hasini@DESKTOP-UTCBO51:~/OSSP/src$ chmod +x test.sh
hasini@DESKTOP-UTCBO51:~/OSSP/src$ ./test.sh
Running OSSP Tests...
Testing memory program:
Data: Hello
Memory released successfully
Testing process program:
Process ID: 879
Parent Process ID: 877
Testing tokenize program:
Enter a command: Tokens:
ls
file.txt
All tests completed.
hasini@DESKTOP-UTCBO51:~/OSSP/src$
```

```
hasini@DESKTOP-UTCQBQ51: ~ × + v
GNU nano 8.7.1 test.sh *
#!/bin/bash

echo "Running OSSP Tests..."

echo "Testing memory program:"
../bin/memory

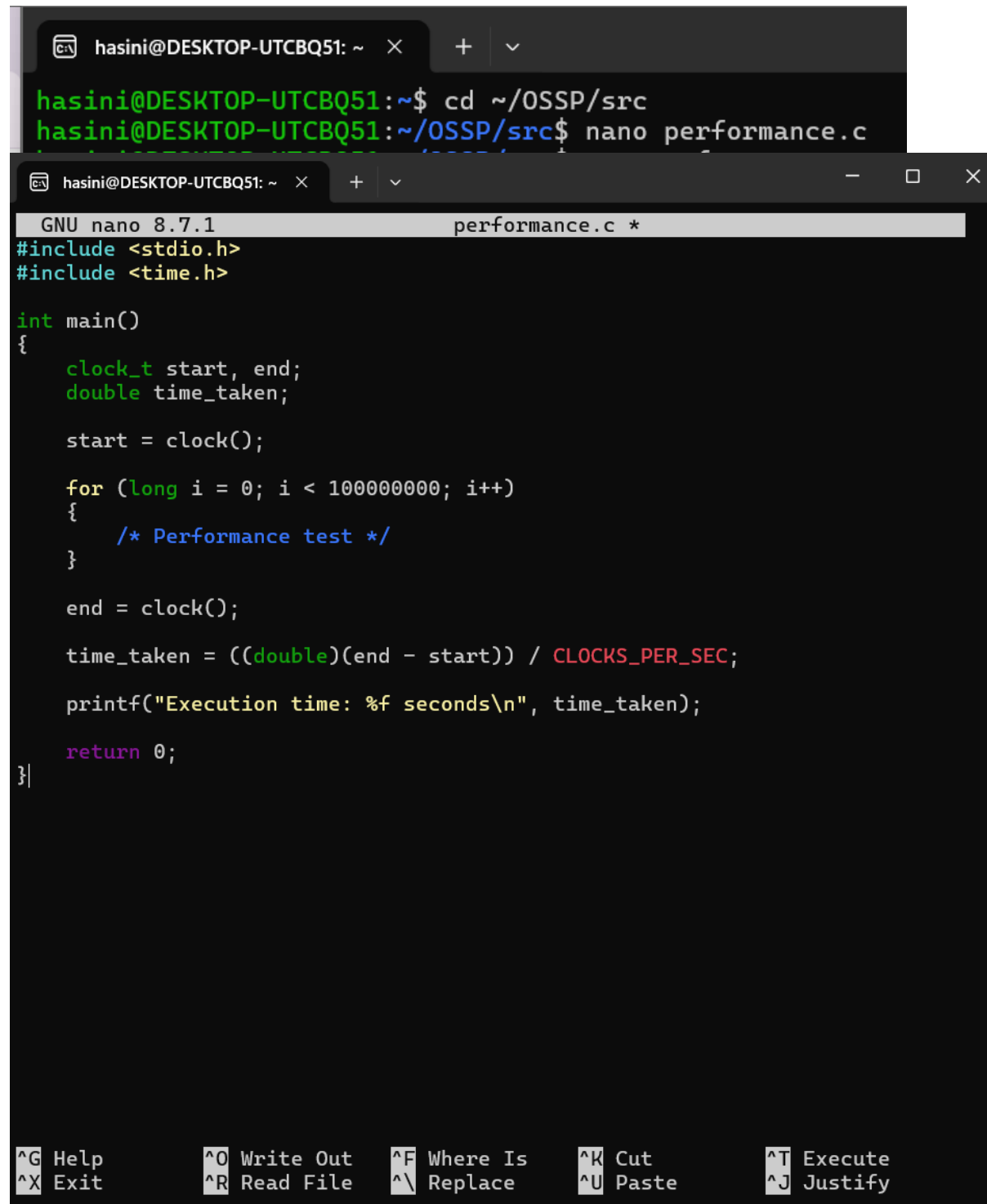
echo "Testing process program:"
../bin/process

echo "Testing tokenize program:"
echo "ls file.txt" | ../bin/tokenize

echo "All tests completed."|

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

23.



The screenshot shows a terminal window with two panes. The top pane shows the user navigating to the directory `~/OSSP/src` and opening the file `performance.c` using the `nano` editor. The bottom pane shows the contents of `performance.c`, which is a C program designed to measure execution time. The program includes `<stdio.h>` and `<time.h>`, and uses `clock()` to measure time. It runs a loop of 10,000,000,000 iterations, each containing a performance test. The execution time is calculated in seconds and printed to the console. The nano editor's status bar at the bottom indicates it is GNU nano 8.7.1 editing `performance.c`.

```
hasini@DESKTOP-UTCBO51: ~$ cd ~/OSSP/src
hasini@DESKTOP-UTCBO51: ~/OSSP/src$ nano performance.c

GNU nano 8.7.1 performance.c *
#include <stdio.h>
#include <time.h>

int main()
{
    clock_t start, end;
    double time_taken;

    start = clock();

    for (long i = 0; i < 1000000000; i++)
    {
        /* Performance test */
    }

    end = clock();

    time_taken = ((double)(end - start)) / CLOCKS_PER_SEC;

    printf("Execution time: %f seconds\n", time_taken);

    return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute  
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

24.

```
hasini@DESKTOP-UTCBO51: ~  
hasini@DESKTOP-UTCBO51:~/OSSP$ find . -maxdepth 2 -type f | sort  
./bin/append  
./bin/background  
./bin/builtin  
./bin/builtins2  
./bin/complex  
./bin/escape  
./bin/fork  
./bin/history  
./bin/input  
./bin/jobs  
./bin/memory  
./bin/modules  
./bin/parse  
./bin/performance  
./bin/pipeline  
./bin/pipes  
./bin/process  
./bin/processgroup  
./bin/quotes  
./bin/redirect  
./bin/signalhandler  
./bin/signals  
./bin/stderr  
./bin/tokenize  
./bin/variables  
./bin/waitpath  
./src/append.c  
./src/background.c  
./src/builtin.c  
./src/builtins2.c  
./src/complex.c  
./src/error.txt  
./src/escape.c  
./src/fork.c  
./src/history.c  
./src/input.c  
./src/jobs.c  
./src/jobs.c.save  
./src/memory.c
```

```
hasini@DESKTOP-UTC BQ51: ~ × + ▾
./src/waitpath.c
hasini@DESKTOP-UTC BQ51:~/OSSP$ git status
fatal: not a git repository (or any of the parent directories): .git
hasini@DESKTOP-UTC BQ51:~/OSSP$ git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warni
ng,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command
:
hint:   git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName fal
se"
Initialized empty Git repository in /home/hasini/OSSP/.git/
hasini@DESKTOP-UTC BQ51:~/OSSP$ git add .
hasini@DESKTOP-UTC BQ51:~/OSSP$ git commit -m "Completed OSSP lab exercises"
Author identity unknown

*** Please tell me who you are.

Run

    git config --global user.email "you@example.com"
    git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: empty ident name (for <hasini@DESKTOP-UTC BQ51.localdomain>) not allow
ed
hasini@DESKTOP-UTC BQ51:~/OSSP$ git config --global user.name "Hasini Reddy"
hasini@DESKTOP-UTC BQ51:~/OSSP$ git config --global user.email "YOUR_EMAIL@ex
ample.com"
```

```
hasini@DESKTOP-UTCBO51: ~ x + v
hasini@DESKTOP-UTCBO51:~/OSSP$ git config --global user.name
git config --global user.email
Hasini Reddy
YOUR_EMAIL@example.com
hasini@DESKTOP-UTCBO51:~/OSSP$ git commit -m "Completed OSSP lab exercises"
[master (root-commit) 802094d] Completed OSSP lab exercises
56 files changed, 886 insertions(+)
create mode 100755 bin/append
create mode 100755 bin/background
create mode 100755 bin/builtin
create mode 100755 bin/builtins2
create mode 100755 bin/complex
create mode 100755 bin/escape
create mode 100755 bin/fork
create mode 100755 bin/history
create mode 100755 bin/input
create mode 100755 bin/jobs
create mode 100755 bin/memory
create mode 100755 bin/modules
create mode 100755 bin/parse
create mode 100755 bin/performance
create mode 100755 bin/pipeline
create mode 100755 bin/pipes
create mode 100755 bin/process
create mode 100755 bin/processgroup
create mode 100755 bin/quotes
create mode 100755 bin/redirect
create mode 100755 bin/signalhandler
create mode 100755 bin/signals
create mode 100755 bin/stderr
create mode 100755 bin/tokenize
create mode 100755 bin/variables
create mode 100755 bin/waitpath
create mode 100644 src/append.c
create mode 100644 src/background.c
create mode 100644 src/builtin.c
create mode 100644 src/builtins2.c
create mode 100644 src/complex.c
create mode 100644 src/error.txt
create mode 100644 src/escape.c
create mode 100644 src/fork.c
```

```
create mode 100644 src/waitpath.c
hasini@DESKTOP-UTCBO51:~/OSSP$ git status
On branch master
nothing to commit, working tree clean
hasini@DESKTOP-UTCBO51:~/OSSP$ On branch master
nothing to commit, working tree clean
On: command not found
nothing: command not found
hasini@DESKTOP-UTCBO51:~/OSSP$ |
```